B. Tech. (Computer Engineering) V Semester Examination 2023-24

OOPS Lab. (CEN-591)

MM: 10 **Time: 1 hour**

3. Define and implement a class hierarchy with base class **Property** with attributes **latitude** (double) and **longitude** (**double**) and pure virtual functions **print** and **sell** and 3 derived classes **House**, Hotel and **Park**. Each of these should have an attribute **sold (boolean)** that indicates whether the property has been sold and **name** (**string**). Define and implement a class **GovtRegistry** that has a single static function called **RegisterSale** that records the name of the property being sold and a function **DisplayRegistry**. Write a driver function that asks the user to enter a choice and then instantiate the appropriate object depending on the user's choice. Get input for 8 objects and store all of the objects in an array (use an array of pointers). Then randomly generate 4 numbers and sell those properties. Print out the final list of the objects and the state of the registry. (CO1, CO3)

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4. Define and implement classes **Point** and **Line**. Point contains two coordinates (**double**) and Line contains two endpoints (**Point**). Overload the operator + inside Point that takes two **Point** operands and returns a **Line** with those as endpoints. Overload the operator * inside **Line** that takes a **Line** and a scale value (double between 0 and 1) and returns a **Point** between the two endpoints such that if the scale value is s then you get the point s * leftEndPt + (1-s) * rightEndPoint as the result. So if the scale value is 0.5, you get the midpoint.

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2. Define and implement a class called **PointND** with attributes **dataPtr** (pointer to type **double**) and **dimension** (unsigned int) that should dynamically allocate memory upon instantiation and release upon destruction. Provide appropriate getters, setters, constructors and destructors. Overload the stream insertion << operator for this class that prints out the L2 (Euclidean) norm and the number of **dimensions**. Overload the addition + operator that returns a new PointND whose dimensions are the sum of the dimensions of the operands and data is the concatenated data from them. (CO1, CO2)

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1. Define and implement a C++ class hierarchy with a base class called **Citizen** with attributes **fullName** (string), **address** (string), **age** (**unsigned int**), 3 virtual functions called **display, whoAmI** and **applyToEnroll**. The derived class is **Voter** with a single attribute **voterID** of type **string** with *exactly* 16 characters. Define and implement a class called **ElectionCommission** with a single static member function called **processEnrollmentApp** that accepts a pointer to a **Citizen** object and returns a **Voter** object if age is 18 or above and the applicant is a **Citizen** and simply returns back the same object if the applicant is a **Voter**. Generate a random **voterID**. Implement and override **display, whoAmI** and **applyToEnroll** functions with appropriate behavior. Provide appropriate constructors and destructors. Write a driver main function to test and demonstrate the functionality of this class hierarchy. (CO1, CO3)